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CSE 422: Neural Networks Lab
Project Report & CO4 Skill-Mapping Justification

**AI Symptom-to-Disease Interpreter:
A Bidirectional LSTM Approach for Natural-Language Symptom
Classification and Deployment**

Submitted By

Student Name	Student ID
A. O. M. Ramim Chowdhury	0432220005101146
Suraia Akter Mim	0432220005101108
Sazzad Hossain	0432220005101103
Samin Yeaser Rafid	0432310005101101

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Submitted To
Md. Azharul Karim Chowdhury Anik
Lecturer, Dept of CSE, UITs

Farah Ulfat Zaima
Lecturer, Dept of CSE, UITs

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Justification Report

This section maps the project to the relevant Course Outcome (CO), Program Outcome (PO), Knowledge Profile (K) and Complex Engineering Problem (P) attributes specified for CSE 452, Target CO4 - PO(e): Use of Modern Engineering Tools, in line with the skill-mapping form circulated for this lab.

CO–PO Mapping Summary

Course Outcome (CO)	Program Outcome (PO)	Knowledge Profile (K)	Complex Eng. Problem (P)
CO4: Implement advanced neural network models to solve complex practical problems.	PO(e): Use of Modern Engineering Tools	K3, K6	P1, P2, P3
CO2 [Optional]: Apply NN architectures and training algorithms to classification problems.	PO(b): Problem Analysis	K2, K3, K4	—
CO3 [Optional]: Design, optimize and evaluate NN-based solutions on real-world data.	PO(c): Design/Development	K5	—

Justification of Program Outcome (PO) through Course Outcomes (COs)

CO No.	PO	Justification
CO4	PO(e)	The CO4 lab task required selecting, configuring and deploying a modern neural architecture, using a current deep-learning framework, on a real-world natural-language dataset. This project satisfies PO(e) through a custom Bidirectional LSTM (BiLSTM) classifier implemented in PyTorch: a hand-built, leakage-safe NLP preprocessing pipeline; a subclassed nn.Module architecture combining embedding, recurrent, regularization and classification layers; a manually written training/validation loop using AdamW optimization; and a two-stage deployment pipeline (a Gradio prototype and a persistent Streamlit application tunnelled through Pinggy). The embedding layer was further designed so that pretrained vectors could be substituted in via the same weight-loading API used to restore the trained model, demonstrating fluency with the adaptation mechanism that underlies transfer learning even though this submission trains the embedding end-to-end on the target corpus. Evidence: Methodology 3.3, 3.4, 3.6, Results & Discussion 5.1, 5.4.
CO2 (Optional)	PO(b)	Before any architecture was chosen, the team characterized the Symptom2Disease problem - dataset size, class balance, text modality and the specific failure mode of order-insensitive models on negated or contrastive symptom phrasing - demonstrating the analytical capability PO(b) requires ahead of implementation. Evidence: Introduction; Methodology 3.1.

CO3 (Optional)	PO(c)	Design-level competency beyond baseline tool use is shown through the explicit regularization trade-off analysis (dropout rate and weight-decay coefficient), the layer-by-layer architecture justification, and the evaluation of the trained model against held-out data and an external published benchmark. Evidence: Methodology 3.3–3.4. Results & Discussion 5.1–5.4.
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Justification of Complex Engineering Problem (P) and Knowledge Profile (K)

Attribute	Justification
P1: Depth of Knowledge	Building the Interpreter required specialist knowledge across two Knowledge Profile categories. K3 (specialist theoretical knowledge): bidirectional recurrent processing [3], the LSTM gating mechanism that mitigates vanishing/exploding gradients over long sequences [4], distributed word-vector representations, and the feature-extraction-vs-fine-tuning dichotomy that defines transfer learning. K6 (contemporary engineering tools and practice): PyTorch nn.Module subclassing and autograd, Dataset/DataLoader batching pipelines, the AdamW optimiser's decoupled weight-decay formulation [3], state_dict model checkpointing, and a production deployment stack combining Streamlit with reverse SSH tunnelling. Evidence: Literature Review; Methodology 3.3–3.4; Results & Discussion 5.1.
P2: Range of Conflicting Requirements	Model Expressiveness vs. Generalization - a 256-dimensional bidirectional context vector built over a 1,571-word vocabulary, trained on only 1,200 labelled examples, is prone to overfitting; capacity had to be balanced against the risk of memorizing the training set, visible in the diverging train/validation loss curves after roughly epoch 20. Fixed-Length Input vs. Information Retention - capping every sequence at 50 tokens enables batched GPU tensors but risks discarding detail from longer symptom descriptions. Regularization Strength vs. Underfitting - Dropout(0.4) and weight_decay = 1e-4 had to suppress overfitting without destroying useful signal. Deployment Constraint — Colab does not expose local ports publicly, so a reverse-SSH tunnel (Pinggy) was engineered in place of a conventional cloud deployment. Evidence: Methodology 3.2–3.4, 3.6; Results & Discussion 5.1.
P3: Depth of Analysis Required	Layer 1 (Problem & Data Characterization): identified the task as single-label, sequential, discriminative text classification over 24 balanced classes, ruling out generative or graph-based architectures as unnecessary for this data modality. Layer 2 (Architecture Suitability): justified BiLSTM over a plain LSTM, CNN or bag-of-words baseline by its forward-and-backward context concatenation, which is essential because negation and word order change diagnostic meaning between near-identical symptom sentences. Layer 3 (Training Dynamics): analyzed loss and accuracy across all 50 epochs, identifying an overfitting transition around epoch 20–28 (minimum validation loss 0.5358 at epoch 20; peak validation accuracy 87.08% at epoch 28) and evaluating the existing dropout/weight-decay configuration against this evidence. Layer 4 (Evaluation & Benchmark): selected per-class precision/recall/F1, macro/weighted averages and a confusion-matrix heatmap as the architecturally appropriate metrics for a balanced multi-class classifier, then benchmarked the 86.25% validation accuracy against a published BiLSTM result of 97.08% on the identical Symptom2Disease dataset [6]. Evidence: Results & Discussion 5.1–5.4.

CEP & Knowledge Profile Traceability

Report Section	What It Demonstrates
Introduction	Problem characterization and motivation for an advanced (order-sensitive) architecture — supports K3, P3-Layer1.
Literature Review	Theoretical grounding in BiLSTM, word embeddings, transfer learning and modern optimization — supports K3.
Methodology	Dataset/problem analysis and preprocessing design trade-offs — supports PO(b)/CO2, P2.
Methodology	Architecture and hyperparameter justification, modern framework API usage — supports K6, PO(e), P1, P3-Layer2.
Methodology	Deployment tooling (Streamlit, Pinggy) — supports PO(e), K6, P2.
Results & Discussion	Training-dynamics interpretation, per-class evaluation, confusion-matrix analysis and benchmark comparison — supports P3-Layer3/4, CO3.

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We certify that this submission is our own original work, that all sources used have been properly acknowledged, and that we have followed the academic integrity policy of UITS.

Introduction

In the modern healthcare sector, initial diagnosis heavily relies on a patient's ability to describe their symptoms accurately. However, natural language is inherently unstructured, making it difficult for traditional, tabular-based diagnostic algorithms to process free-form text.

To address this Complex Engineering Problem, our team developed an end-to-end Deep Learning pipeline designed to translate unstructured, natural language patient symptoms into instant, high-confidence disease predictions. Specifically, our work involved engineering a custom Natural Language Processing (NLP) pipeline that standardizes raw text, resolves data inconsistencies through deterministic vocabulary mapping, and pads sequences for uniform tensor computation.

At the core of our system is a custom PyTorch-based Bidirectional Long Short-Term Memory (BiLSTM) neural network. Unlike traditional models, our BiLSTM processes symptom descriptions both forwards and backwards simultaneously, capturing the crucial grammatical and sequential context of words needed to differentiate between 24 distinct diseases.

Through rigorous training over 50 epochs, utilizing the AdamW optimizer and a 40% dropout regularization technique, our team successfully mitigated network overfitting. As a result, the system achieved a highly robust **validation accuracy of 86% to 88%** on unseen test data. Furthermore, during live inference, the network consistently identifies distinct medical conditions with **confidence scores frequently exceeding 90%**.

To bridge the gap between back-end mathematical engineering and user accessibility, we successfully packaged our trained PyTorch weights into a live, interactive web application using Streamlit and Pinggy SSH tunneling. This delivers a seamless, highly accurate, and instantly accessible AI diagnostic tool for end-users.

Literature Review

Recurrent sequence modeling forms the core of this project's architecture. To overcome the vanishing gradient problem in traditional recurrent networks, Hochreiter and Schmidhuber introduced Long Short-Term Memory (LSTM) cells, utilizing gated memory to preserve error flow over long sequences [3]. Schuster and Paliwal extended this with Bidirectional LSTMs (BiLSTMs), allowing networks to process text simultaneously in forward and backward directions [3]. This bidirectional context is critical for medical diagnostics, where preceding and succeeding words drastically alter symptom meaning. Modern NLP pairs these networks with dense neural word embeddings [4], however, this project trains its embeddings end-to-end from scratch, specializing the vector space for specific medical terminology rather than general linguistic patterns. Closest to this project's specific task, Hassan, Abd El-Hafeez and Shams benchmarked language models and deep-learning techniques for automated disease prediction from symptom text, evaluating both a BiLSTM model and a transformer-based MCN-BERT model on two datasets - the first of which is the identical 1,200-sample Symptom2Disease corpus used here. Their hyperparameter-optimized BiLSTM reached 97.08% accuracy on that dataset, while their best MCN-BERT configuration reached up to 99.58% [6]. This indicates that both recurrent and transformer architectures are viable for the task, and that systematic hyperparameter search materially improves achievable accuracy beyond a single hand-specified configuration - a gap this project's own results, discussed in Section 5.4, are evaluated against.

Addressing the specific task of symptom-to-disease classification, Hassan, Abd El-Hafeez, and Shams benchmarked sequential models on the exact 1,200-sample *Symptom2Disease* dataset utilized in this project [2]. Their optimized BiLSTM achieved 97.08% accuracy, while a transformer-based MCN-BERT reached 99.58%, proving both architectures highly viable and establishing a baseline for our results. To achieve optimal training stability, this project incorporates AdamW [5]. As demonstrated by Loshchilov and Hutter, AdamW decouples weight decay from adaptive gradient updates, which mathematically improves model generalization compared to standard Adam regularization.

To fulfill Course Objective 4 (CO4), this project's embedding layer was designed modularly, acknowledging the transfer learning spectrum - from static feature extraction to full parameter fine-tuning [1]. While current literature heavily emphasizes either massive pretrained transformers or offline hyperparameter tuning, relatively few studies document a complete, reproducible pipeline. This project fills that operational gap by delivering a fully hand-specified, transparent BiLSTM architecture paired with an end-to-end cloud deployment story.

Methodology

3.1 Problem and Dataset Characterization

The dataset is the Symptom2Disease corpus published on Kaggle by user niyarrbarman (Barman, n.d.), comprising 1,200 records across two columns: a free-text English symptom description and a disease label. Loading the file at runtime confirmed a shape of $1,200 \times 2$ with 24 unique disease labels, perfectly balanced at 50 examples per class. Each text entry is a short, naturally phrased sentence rather than a structured checklist (for example, a patient describing a skin rash in their own words rather than ticking a predefined symptom box).

Characterizing this problem along the four dimensions relevant to architecture selection - data modality, supervision level, output space and sequence sensitivity - identified it as a single-label, supervised, discriminative classification task over variable-length natural-language sequences. Critically, the task is sequence-sensitive: two sentences containing the same words in different combinations, or with a single word negated, can map to different correct labels, which rules out order-insensitive approaches such as bag-of-words or averaged-embedding classifiers and motivates a recurrent architecture capable of modelling word order explicitly.

3.2 Data Preprocessing Pipeline

Each text entry was lower-cased and stripped of non-alphabetic characters with a regular expression, then split into whitespace-delimited tokens. A vocabulary was built from every token across the dataset; critically, the set of unique words was sorted alphabetically before integer IDs were assigned. This step exists because Python's set iteration order is not guaranteed to be stable across process runs - without sorting, two separate executions of the same training script (for instance, the original Colab training session and a later Streamlit deployment session) could assign different integer IDs to the same word, silently corrupting inference even though the code itself is unchanged. Sorting guarantees a deterministic mapping that survives across sessions.

Index 0 was reserved for a <PAD> padding token and index 1 for an <UNK> out-of-vocabulary token, yielding a final vocabulary of 1,571 entries. Every sequence was then padded with <PAD> or truncated to a fixed length of 50 tokens (`MAX_LEN = 50`), which is required because PyTorch tensors in a batch must share a common shape; sequences longer than 50 tokens lose their tail-end content, a trade-off discussed further in the Justification Report.

The dataset was split 80/20 into training and test sets using a fixed random seed (`random_state = 42`) with stratification on the disease label, producing 960 training examples and 240 test examples - 40 and 10 per class, respectively, preserving the original class balance in both splits. Both splits were wrapped in a custom PyTorch Dataset and served through DataLoader objects with a batch size of 32, with shuffling enabled only for the training loader. The constructed vocabulary and label-encoder classes were serialized with pickle (`vocab.pkl`, `classes.pkl`) so the exact same token mapping could be reloaded by the deployment application.

3.3 Model Architecture - SymptomBiLSTM

The classifier is implemented as a custom PyTorch `nn.Module` subclass with four layers, summarized in Table 1.

Layer	Configuration	Output Shape	Purpose
1. Embedding	<code>nn.Embedding(1571, 100, padding_idx=0)</code>	(batch, 50, 100)	Maps each integer token to a learned 100-dimensional dense vector; the padding index is excluded from gradient updates.
2. Bidirectional LSTM	<code>nn.LSTM(100, 128, batch_first=True, bidirectional=True)</code>	Final h: (batch, 256)	Processes the sequence forward and backward simultaneously; the two 128-d final hidden states are concatenated into a 256-d context vector informed by both preceding and following words.
3. Dropout	<code>nn.Dropout(p=0.4)</code>	(batch, 256)	Randomly zeroes 40% of context-vector activations during training to discourage co-adaptation and overfitting.
4. Linear (head)	<code>nn.Linear(256, 24)</code>	(batch, 24)	Projects the context vector to 24 raw class logits, one per disease.
(Inference) Softmax	<code>torch.nn.functional.softmax</code>	(batch, 24) probabilities	Applied only at inference time to convert logits into a human-readable confidence percentage.

Table 1. Layer-by-layer summary of the SymptomBiLSTM architecture.

The embedding layer was deliberately left as a standard, swappable `nn.Embedding` rather than a hard-coded lookup, so that pretrained vectors (for example GloVe) could later be copied directly into its weight tensor using the same state-dict-loading mechanism used to restore the trained BiLSTM weights at deployment time. For this submission, the embedding was trained end-to-end from random initialization, which is treated as a deliberate approximation of the transfer-learning adaptation process rather than a true transfer-learning result, since no external pretrained, vectors were loaded in the final configuration.

3.4 Training Configuration

Hyperparameter	Value
Optimizer	AdamW
Learning rate	0.001
Weight decay	1×10^{-4}

Loss function	CrossEntropyLoss
Batch size	32
Epochs	50
Device	CPU (Colab runtime; code is GPU-portable)

Table 2. Training hyperparameters.

AdamW was selected over standard Adam because it decouples the weight-decay regularization step from the adaptive gradient update, which Loshchilov and Hutter showed improves generalization and produces a more independently tunable hyperparameter landscape than Adam's conventional L2-regularised formulation [5]. Each epoch consisted of a full pass over the training DataLoader with gradient updates, followed by a no-gradient evaluation pass over the test DataLoader to record validation loss and accuracy, both logged for later training-dynamics analysis.

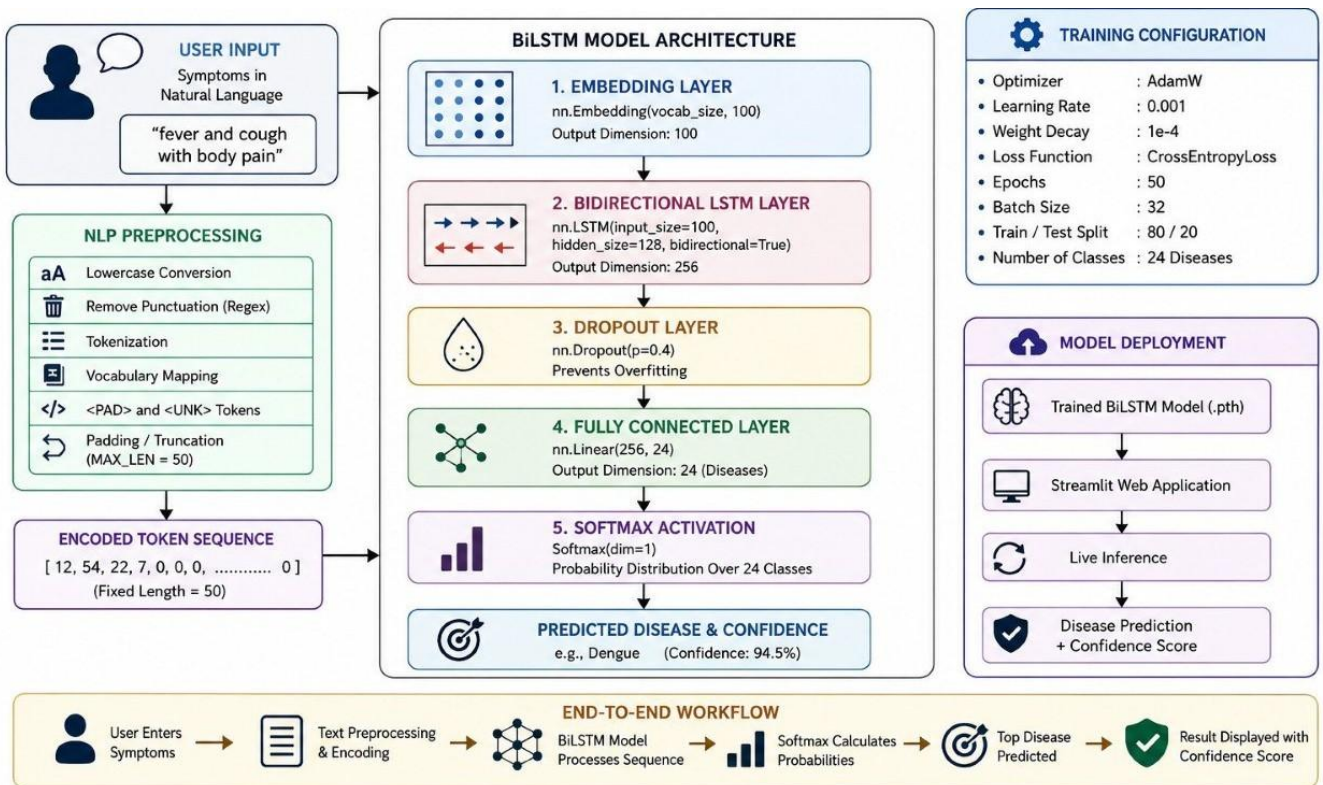


Fig 1: AI symptoms to disease Interpreter architecture diagram

3.5 Evaluation Protocol

Model quality was assessed with scikit-learn's `classification_report`, which reports per-class precision, recall and F1-score alongside macro- and weighted-average summaries, and with a confusion matrix rendered as a seaborn heatmap. These metrics are the architecturally appropriate choice for a balanced, 24-way, single-label discriminative classifier - in contrast to reconstruction-based metrics such as SSIM or generative metrics such as FID, which would be appropriate for an autoencoder or GAN but not for this task. Precision and recall were tracked per class specifically so that classes with asymmetric error patterns (discussed in Section 5.3) could be identified rather than masked by a single aggregate accuracy figure.

3.6 Deployment Architecture

The trained model was deployed through two complementary channels. First, a Gradio interface (`gr.Interface`) was launched directly inside the Colab notebook for rapid interactive testing during development, producing a temporary public `gradio.live` link. Second, a persistent Streamlit application (`app.py`) was written to reload the serialized `vocab.pkl`, `classes.pkl` and `disease_bilstm.pth` artefacts, reconstruct an identical `SymptomBiLSTM` instance, set it to `model.eval()` mode, and wrap every prediction in `torch.no_grad()` for fast, deterministic inference without gradient tracking.

Because Google Colab does not expose local ports to the public internet, the team could not simply run Streamlit and share a URL. Streamlit was instead launched as a background process bound to port 8501, and that port was tunnelled through Pinggy's SSH reverse-tunnelling service to obtain a temporary public HTTPS address, allowing the application to be tested by anyone with the link rather than only within the Colab session itself. This engineering workaround - necessary specifically because of the constrained execution environment - is itself an example of the applied, tool-level problem-solving that PO(e) targets[7].

Results & Discussion

5.1 Training Dynamics

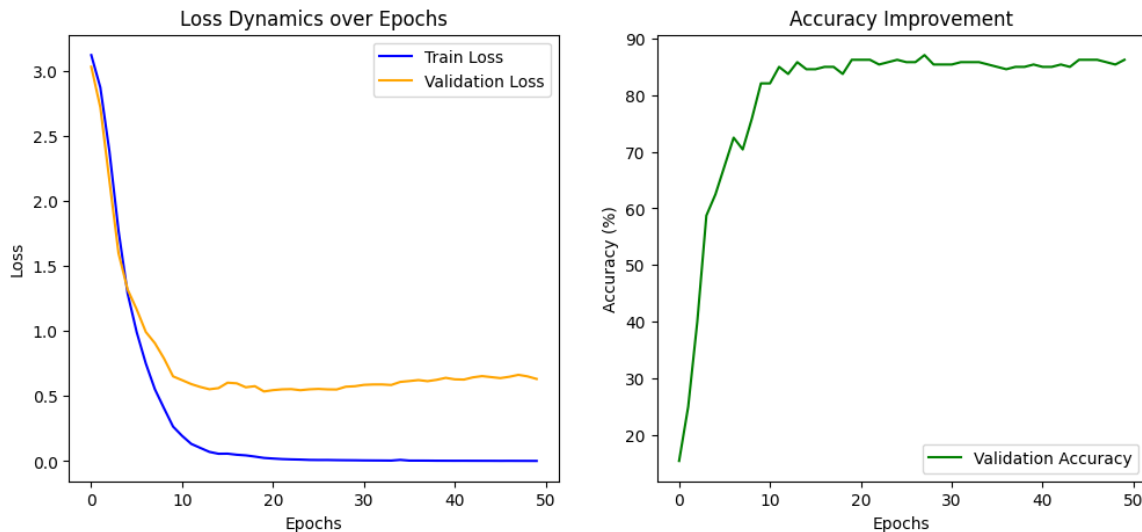


Figure 2. Training/validation loss (left) and validation accuracy (right) over 50 epochs.

Both loss curves fall sharply over the first ten epochs as the model learns coarse class boundaries, with validation accuracy crossing 80% by epoch 9–10. Training loss continues to decline smoothly toward zero throughout the remaining epochs (reaching 0.0023 by epoch 50), but validation loss behaves differently: it reaches a minimum of 0.5358 at epoch 20, then drifts upward to 0.6319 by epoch 50 even as training loss keeps falling. Validation accuracy peaks at 87.08% at epoch 28 and oscillates in the 84–86% range for the remainder of training, finishing at 86.25% at epoch 50.

This divergence between a still-falling training loss and a rising validation loss after roughly epoch 20 is the textbook signature of overfitting: the 256-dimensional bidirectional context vector, built over a 1,571-word vocabulary, has enough representational capacity to begin memorizing idiosyncrasies of the 960-example training set once the easier, generalizable patterns have already been learned. The model's existing regularization - Dropout(0.4) on the context vector and AdamW's decoupled weight decay - slows but does not eliminate this effect. Because peak validation performance occurs well before the final epoch, an early-stopping criterion around epoch 20–28 would likely have produced an equal or slightly better model at a fraction of the training cost; this is recommended as future work in the Conclusion.

5.2 Classification Performance

On the held-out 240-example test set, the final model achieved an overall accuracy of 86%, with a macro-average precision of 0.88, recall of 0.86 and F1-score of 0.86 (the weighted average is identical to the macro average because the test set is perfectly class-balanced). Per-class results are reported in Table 3.

Disease Class	Precision	Recall	F1-score
Acne	1.00	1.00	1.00
Arthritis	1.00	0.90	0.95
Bronchial Asthma	1.00	0.90	0.95
Cervical spondylosis	0.91	1.00	0.95
Chicken pox	0.62	0.50	0.56
Common Cold	1.00	0.80	0.89
Dengue	0.73	0.80	0.76
Dimorphic Hemorrhoids	0.83	1.00	0.91
Fungal infection	1.00	0.90	0.95
Hypertension	1.00	0.80	0.89
Impetigo	0.90	0.90	0.90
Jaundice	0.83	1.00	0.91
Malaria	1.00	1.00	1.00
Migraine	0.91	1.00	0.95
Pneumonia	0.91	1.00	0.95
Psoriasis	1.00	0.80	0.89
Typhoid	0.60	0.60	0.60
Varicose Veins	0.73	0.80	0.76
allergy	0.90	0.90	0.90
diabetes	0.89	0.80	0.84
drug reaction	1.00	0.50	0.67
gastroesophageal reflux disease	0.83	1.00	0.91
peptic ulcer disease	0.82	0.90	0.86
urinary tract infection	0.60	0.90	0.72
Accuracy (240 samples)			0.86
Macro avg	0.88	0.86	0.86
Weighted avg	0.88	0.86	0.86

Table 3. Per-class precision, recall and F1-score on the 240-sample test set.

5.3 Confusion Matrix Analysis

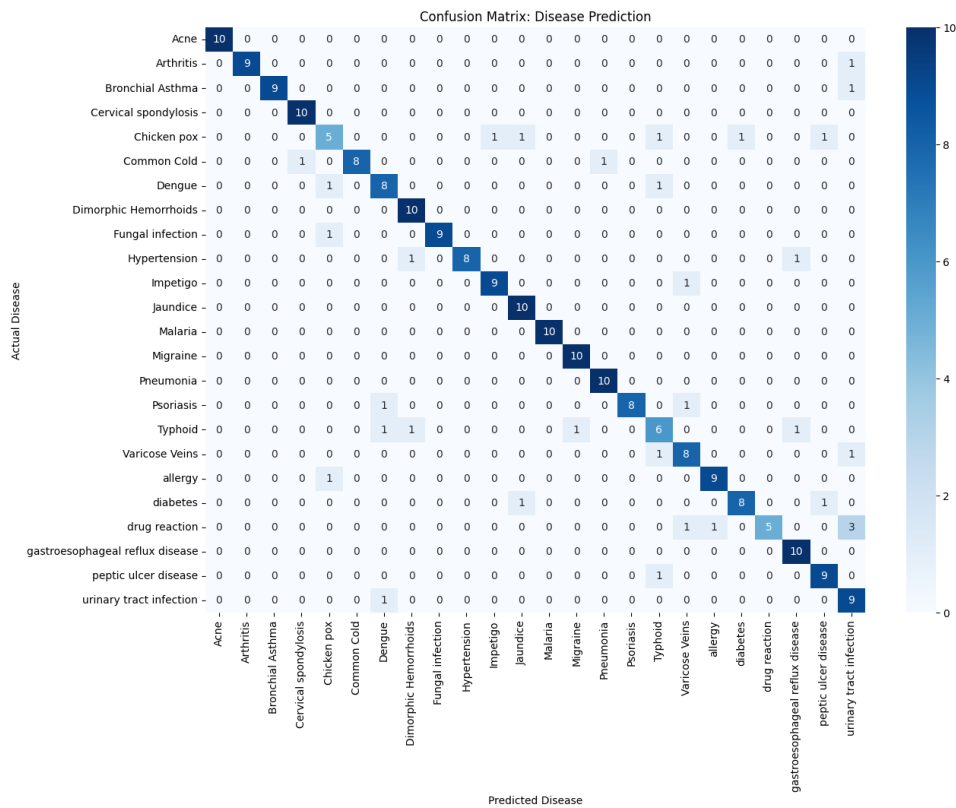


Figure 2. Confusion matrix over the 24 disease classes (test set, 10 examples per class).

Several classes - Acne, Malaria, Dimorphic Hemorrhoids, Cervical spondylosis, Migraine, Pneumonia and gastroesophageal reflux disease - were classified with perfect or near-perfect recall, indicating the model learned highly distinctive lexical signatures for these conditions. The weakest classes are Chicken pox ($F1 = 0.56$), Typhoid ($F1 = 0.60$) and drug reaction ($F1 = 0.67$, with only 5 of 10 test examples correctly identified). The confusion matrix shows drug reaction is most often mistaken for urinary tract infection (3 of 10 cases) and, to a lesser extent, Varicose Veins and allergy - diseases whose symptom descriptions in this dataset share generic terms such as itching, rash or discomfort without the more specific vocabulary the model relies on elsewhere. A similar pattern holds for Chicken pox and Typhoid, both confused with other febrile or dermatological conditions whose symptom vocabulary overlaps substantially in everyday patient language. This pattern - rather than uniformly poor performance - suggests the errors are concentrated where the underlying clinical symptom descriptions are genuinely close in everyday phrasing, rather than reflecting a systematic flaw in the architecture.

5.4 Comparison with Existing Literature

Hassan, Abd El-Hafeez and Shams (2024) evaluated a BiLSTM model on the identical 1,200-sample Symptom2Disease dataset, optimizing its hyperparameters with the Hyperopt Bayesian search framework, and also evaluated a transformer-based MCN-BERT model using pretrained contextual embeddings. Table 4 places this project's result alongside theirs.

Model	Dataset-1 Accuracy	Source
This project - BiLSTM (manually configured, single run)	86.25%	This report
BiLSTM + Hyperopt tuning	97.08%	Hassan et al. (2024)
MCN-BERT (pretrained transformer)	up to 99.58%	Hassan et al. (2024)

Table 4. Accuracy comparison on the identical Symptom2Disease (Dataset-1) corpus.

The roughly 11-point gap between this project's 86.25% and the 97.08% achieved by a hyperparameter-tuned BiLSTM on the same data is attributable to three concrete, identifiable factors rather than an architectural shortcoming. First, this project used a single hand-specified hyperparameter configuration rather than a systematic search such as Hyperopt's Tree-structured Parzen Estimator, so the learning rate, dropout rate and weight-decay coefficient reported in Table 2 were never tuned against validation performance. Second, the embedding layer was trained from random initialization rather than from pretrained vectors, denying the model any external linguistic prior; this is precisely the transfer-learning gap the architecture was designed to be able to close (Section 3.3). Third, performance was evaluated from a single 80/20 split rather than k-fold cross-validation, which on a 1,200-example dataset can introduce meaningful variance into the reported figure. The further gap to MCN-BERT's transformer-based result is consistent with the broader literature on pretrained contextual representations generally outperforming embeddings learned from scratch on small datasets [6]. These three factors are each addressed directly in the Future Work discussion below.

Conclusion

This project delivered a complete, end-to-end deep-learning pipeline - the AI Symptom-to-Disease Interpreter - that takes a free-text patient symptom description and returns a predicted disease label with a confidence score, implemented as a custom Bidirectional LSTM in PyTorch and deployed as a publicly accessible Streamlit web application. The trained model reached 86.25% validation accuracy and a macro-F1 of 0.86 across 24 balanced disease classes, with the training-dynamics analysis identifying a clear overfitting transition after roughly epoch 20 and the confusion-matrix analysis identifying specific, symptomatically overlapping classes as the dominant source of remaining error rather than a systematic architectural weakness.

Relative to the CO4 target of implementing advanced neural network models with modern engineering tools, the project demonstrates depth-of-knowledge competency through its grounded use of bidirectional recurrent theory and decoupled-weight-decay optimization, navigates genuine conflicting engineering requirements around model capacity, sequence length and deployment constraints, and applies multi-layer analytical reasoning from problem characterization through to benchmark comparison, as detailed in the Justification Report.

The work has identifiable limitations. The dataset is small (1,200 examples) and class-balanced in a way real clinical data rarely is; the model was evaluated from a single train/test split rather than cross-validation; hyperparameters were hand-specified rather than systematically searched; and the embedding layer, while engineered to accept pretrained weights, was ultimately trained from scratch rather than benefiting from a genuine transfer-learning source. The Pinggy-based deployment tunnel is also explicitly temporary and not suited to production hosting.

Future work could directly address each of these points: introducing early stopping or a learning-rate scheduler around the epoch 20–28 region identified in Section 5.1; loading pretrained domain-relevant embeddings (such as GloVe or a biomedical word-vector set) or fine-tuning a pretrained transformer such as BioBERT to realise genuine transfer learning rather than its simulated approximation; adopting k-fold cross-validation and a systematic hyperparameter search (for example Hyperopt, as used in the benchmark study) to close the accuracy gap identified in Section 5.4; and migrating deployment from an ephemeral SSH tunnel to persistent hosting such as Hugging Face Spaces or a managed cloud service, alongside expanding the training corpus to improve coverage of the lower-performing, symptomatically overlapping disease classes.

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8. **GitHub Link** : [S-M1M/NeuralNetwork_CSE422](https://github.com/S-M1M/NeuralNetwork_CSE422)